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Overcoming 'Diffusion Limits' – Principles required to measure high molar mass polymers by diffusion ordered NMR

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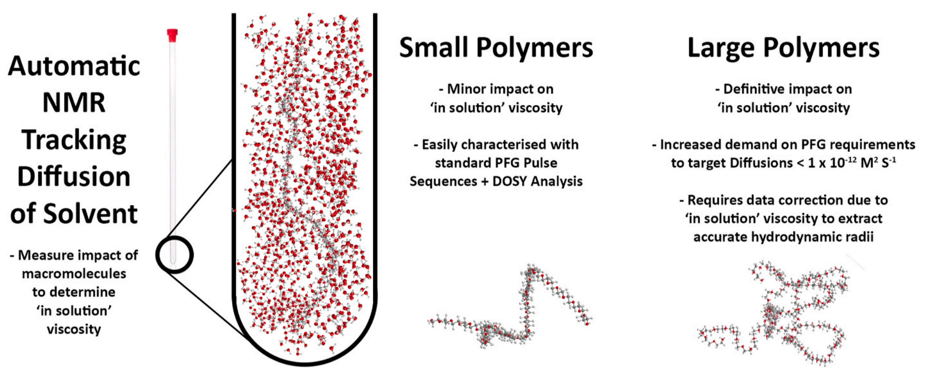
HIGHLIGHTS

- 'In Solution' viscosity can be automatically measured by analysis of solvent diffusion.
- Viscosity correction is necessary during DOSY analysis of large macromolecules.
- Large molar mass polymers require more demanding PFG hardware to extract accurate diffusion data.

GRAPHICAL ABSTRACT



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ABSTRACT

Question: This paper studies the importance of resolving 'in-solution' viscosity to determine an accurate hydrodynamic radii for high molar mass or high dispersity macromolecules via DOSY NMR. Analysis of polymer size via diffusion NMR has become increasingly more common, however as in-solution viscosity increases NMR output becomes more complex and requires dedicated methodologies (both in the instrumentation and data treatment) that can sufficiently resolve slowly diffusing analytes.

Results: Diffusion measurements were used to determine hydrodynamic radii of dissolved polymer chains of materials across a broad molar mass range in multiple solvents. Studied systems included poly(ethylene glycol), poly(ethylene oxide), poly(styrene), poly(methyl methacrylate) and poly(*N*-isopropylacrylamide) and all are shown to match known literature values for dissolved polymer coils with a high degree of accuracy. However, it is shown that it is essential to use the "in-solution viscosity", which can be obtained by applying a viscosity correction factor to the pure solvent viscosity. It was found that % error in outputs correlates to the viscosity of the solvent, with low viscosity solvents contributing to a higher variability in output data. We have also shown how the experimental range of the technique can be expanded to high molar mass (in excess of 1 million g mol^{-1}), or high viscosity, and demonstrated the advantages of a diffusion optimised NMR probe (Bruker DiffBB) to target slowly diffusing chemical species.

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Significance: The presence of even small quantities of large molar mass polymer analytes (2 mg mL^{-1}) has an impact on in-solution viscosity, and thus provides a systematic offset in output diffusion values that are commonly used to interpret polymer sample size. DOSY NMR data include the diffusion of the solvent in-solution. Therefore, DOSY NMR measurements alone, with no internal or external standard besides the solvent itself, can be used to correct for this, allowing for prediction of an accurate hydrodynamic radius (and thus molar mass) of large, slowly diffusing, materials.

1. Introduction

1.1. Context

Diffusion ordered spectroscopy (DOSY) is an established NMR technique providing molecular level size and composition data of polymers, low molar mass compounds and mixtures. It is operated using pulsed field gradient (PFG) sequences, which stimulate signals from target isotopes that decay according to their molecular motion, allowing formation of a 2 dimensional 'map' of the analyte sample [1]. Several PFG processing methods are available but the most commonly applied system is DOSY, first outlined by Morris and Johnson in 1992 [2]. The application of DOSY to polymer analysis was set out by Groves [3], who has pointed to the lack of reporting of the key DOSY NMR parameters for polymers in the literature. Despite the availability of NMR, currently the gold standard for polymer analysis are chromatographic techniques and several advances in recent years have steadily increased the upper limit of detection for the study of large molecules [4,5]. Typically, for size exclusion chromatography the upper and lower limits of detection are considered to be the exclusion limit of the stationary phase [6] but another inherent limitation of very high molar mass polymers via these techniques are shear degradations that occur at high flow rates [7]. As there is no physical separation occurring within DOSY NMR, and also no requirement for shear, the technique appears to be an appealing method of overcoming these barriers in sizing very large polymer systems.

In recent years there has been a sustained effort to use DOSY NMR measurements to gain increased information about the molar mass and size of polymers [8–17] and biological molecules [18], often using an internal standard small molecule [19,20], calibration with polymer standards measured via another technique [9,10,18] or external standards [21] as references. Changes in D have previously been used as a means of measuring R_H[22–27] and NMR is becoming increasingly cost effective, with a rise in the use of DOSY capable benchtop spectrometers [15,28]. Recently DOSY NMR has repeatedly been put forward as a tool for universal molar mass characterisation [12,14–17]. Size exclusion chromatography suffers from limitations due to the exclusion limits based on stationary phase pore size. These limitations are not a feature of diffusion NMR analysis but there are other instrumentational limits that should be acknowledged for method development of the technique. This manuscript attempts to outline these and the solutions in the context of analysing high molar mass polymers.

In this paper we examine both the considerations required for treatment of polymer diffusion data to get accurate polymer sizes of high molar mass materials, and also demonstrate the potential advantages of broadband gradient probes optimised for strong gradient pulses (Bruker DiffBB) with the power to apply extremely high gradient strengths necessary to fully resolve particles with slow diffusions ($>1 \times 10^{-13}$ m^2 s^{-1}). We show how this setup is required to accurately resolve the hydrodynamic radii of dissolved high molar mass polymers. To do this we demonstrate how a single DOSY measurement of a dilute polymer in deuterated solvents alone, with no reference to any other measurement, can infer both the dynamic viscosity of the solution and the hydrodynamic radius of that polymer chain. Doing this we show how simultaneous measurement of solvent-in-solution viscosity is essential to provide a more precise determination of polymer coil size. Solvent-in-solution viscosity is obtained directly from the DOSY experiment and is derived from the self-diffusion of the solvent in the polymer solution. This solvent polymer solution viscosity is different from the pure solvent value.

The approach taken to achieve this is to study solutions of poly(ethylene oxide)/poly(ethylene glycol) (PEO/PEG) of varying molar masses, in multiple solvents, using different instrument conditions. Furthermore, as polymer conformation is solvent dependent, we have compared the same data in different solvent systems to explore additional variables this introduces for accurate analysis. For example in aqueous solution PEG is expected to form semi-ordered structures whilst reverting to a more random conformation in solvents such as chloroform [29]. Additionally poly(styrene) (PS), poly(methyl methacrylate) (PMMA) and poly(*N*-isopropylacrylamide) (PNIPAM) were also studied. The former two polymers are well-studied with defined solution properties[30–41], and furthermore are commonly used as calibration standards for size-exclusion chromatography measurements. PNIPAM is a technologically important polymer that displays temperature dependant phase behaviour in water.

1.2. Theory

Diffusion Ordered NMR Spectroscopy (DOSY) measurements operate via the principal that all molecules in a liquid or solution are under constant motion, and this mobility in a stationary liquid can be considered to equate to a 'self-diffusion' (D) value. Diffusion for spherical particles can be modelled using the Stokes-Einstein equation (1).

$$D = \frac{k_B T}{6\pi\eta r} \tag{1}$$

Here, k_B is the Boltzmann constant, T the temperature, η the dynamic viscosity of the medium through which the spheres diffuse and r is the radius of the spheres. Many molecules, regardless of molecular shape can be modelled as a collection of spheres and so the equation proves to be valid for a wide range of materials, although Avramov showed that this relationship is not applicable to fast moving, small ions [42]. In the field of polymers the radius of these spheres is often considered equivalent to radius of a non-draining polymer coil. This treatment assumes that polymers diffuse as non-draining spheres of average hydrodynamic radius, R_H. η of the medium can be obtained by considering the self-diffusion of the solvent in the polymer solution [43]. The self-diffusion of the solvent is known to vary with polymer concentration and most theoretical models are based around changes in free volume available for solvent to move into [42,44–46]. The presence of the polymer can be considered to reduce the fraction of free volume available such that the self-diffusion coefficients decrease as polymer concentration increases. Several treatments of polymer DOSY data assume that the viscosity in the Stokes-Einstein equation can be estimated as the viscosity of pure solvent [8,9,11]. However, the DOSY experiment has long been known to provide the distribution of self-diffusion constants of both the solvent [47,48] and polymer chain at a defined polymer concentration and the data can be used to provide η that is appropriate at this concentration of polymer and the temperature of the system.

DOSY measurements use the principal that the intensity of signal decay of a pulsed field NMR measurement is attenuated by the diffusion time of the experiment (Δ) and the gradient pulse duration when applying the magnetic field (δ). These terms are interconnected via the Stejskal-Tanner equation (2) [49]. To fully define this equation q

represents the applied gradient strength and γ represents the gyromagnetic ratio. Via series of measurements where g (or, less common, Δ or δ) is progressively varied, the signal intensity (I) is observed by a measurement compared to the reference unattenuated signal (I_o) (i.e. I when q is zero) [50].

$$I = I_o e^{-Dg^2\gamma^2\delta^2\left(\Delta-\frac{\delta}{3}\right)} \tag{2}$$

This equation shows that as the magnetic gradient strength of the field is increased the observed intensity of the proton signal will decrease. If the gradient field increases over an area from which sufficient signal intensity is lost the decay can be fitted allowing calculation of the diffusion (D).

At this point we note that there are different approaches to determining intensity of NMR peaks – using either the maximum peak intensity or the sum area under the peak. Analysis of macromolecules however tends to give broadened peaks, depending on sample-solute interactions. As such, for the purposes of this manuscript we have fitted the peak intensity – so D will reflect the diffusion at the peak ^{1}H intensity (hence D should reflect the number average molar mass (M_n). It is also possible to correlate diffusion values from across the peak and determine an average D value but that has not been applied in this study.

Rearrangement of the Stokes-Einstein equation means the hydrodynamic radius of a polymer can be determined using the diffusion of a polymer via rearrangement of Equation (1) to produce (3).

$$R_H = \frac{k_B T}{6\pi\eta D} \tag{3}$$

Furthermore, for Gaussian chains if the molar mass of the polymer sample (M) is already known, knowledge of the hydrodynamic radius means that the sample intrinsic viscosity can be determined [43] using a rearranged equivalent to the Fox-Flory equation [36] where V_H is the polymer hydrodynamic volume and N = Avogadro's number as shown in (4).

$$[\eta] = \frac{2.5NV_H}{M} \tag{4}$$

The volume of the equivalent sphere of the hydrodynamic radius can be calculated by Equation (5).

$$V_H = \frac{4}{3}\pi R_H{}^3 \tag{5}$$

Knowledge of intrinsic viscosity is a key parameter in polymer characterisation, thus determining it from NMR measurements represents a significant advancement of the technique. The Mark-Houwink-Sakurda (MHS) (Equation (6)) relates intrinsic viscosity information with molar mass and polymer shape.

$$[\eta] = kM_v{}^\alpha \tag{6}$$

Here K and α represent constants that are proportional to the polymers molecular conformation in the solvent, and α would be expected to ≈ 0.5 if the polymer existed in theta conditions (i.e. exists as a random coil with unperturbed dimensions). In this work, as we are discussing narrow molar mass polymer standards, we have assumed M_v can be substituted for M_w for the purposes of data analysis.

The proton signal for the polymer backbone and peaks from the solvent are separated in the DOSY experiment. Ideally there should be no overlap between the solvent proton peak and the species under direct investigation. Importantly because experiments provide the distribution of diffusion constants the moments of [η] and V_H can be obtained from broadly distributed samples.

Although the viscosity of deuterated solvents can differ slightly from proterated solvents at a given temperature [51], the intrinsic viscosity of polymers within deuterated solutions are similar to the proterated analogues below large molar masses [52]. Knowing this a pre-measurement is required to obtain the intrinsic diffusion of the solvent (D_S) in its pure form under the experimental conditions. From this we can then infer that if the viscosity of the sample changes the diffusion of the solvent proton peaks will shift accordingly.

Several different groups have previously published ways of accounting for the solvent diffusion. Early studies using DOSY to estimate molar mass of dilute samples simply indicated a visual check to ensure that the diffusion of the solvent did not significantly differ from the base solvent – therefore proving no correction factor was required and dismissing data where it was [18,53]. This is problematic and indicates confusion within some of the scientific community; the η term does not indicate the 'solvent' viscosity but the 'solution' viscosity. Correcting for viscosity changes has been shown to provide largely accurate data for small to medium sized polymer systems in a range of solvents [11,12,16]. These examples of recent work predominately achieve this by separating out the contributing components of solvent diffusion and then accounting for different solvent viscosities to produce solvent independent calibration plots.

In 2017 Swift et al. presented one option where the known inverse relationship between viscosity and diffusion could be used to estimate changes in solution viscosity using Equation (7). Here we compare how, as the diffusion of a molecule is compared across two solution environments (1S and 2S) it will reflect changes in its surrounding viscosity.

$$D_{1S}\eta_{1s} = D_{2S}\eta_{2S} \tag{7}$$

Therefore, knowledge of η and D of the pure solvent (η_{1S} and D_{1s}) and D of the solvent in solution (D_{2s}) can be used to extract η of the solvent in solution. This equation holds because conceptually in dilute solutions the hydrodynamic volume of any species is considered to be concentration independent. Therefore at two separate low concentrations, where the hydrodynamic radii of C_1 and C_2 are compared, the diffusion and dynamic viscosity components must remain equivalent and Equation (3) can be rearranged to form Equation (8).

$$R_H = \frac{k_B T}{6\pi\eta_{1s}D_1} = \frac{k_B T}{6\pi\eta_{2s}D_2} \tag{8}$$

Note that η is the viscosity of the solvent in solution. Furthermore, it has been shown that the product $D\eta$/T is constant for the self-diffusion of small molecule solvents [42], and Hiller and Grabe have shown that there is universality of these equations in that they apply across multiple solvent systems [17].

A similar approach has been adopted by other research groups using an independent internal standard, such as tetramethylsilane (TMS), which has historically been used as a reference standard for calibrating NMR shifts [20,54,55]. In recent years it has been shown that variations in TMS diffusion reflect viscosity shifts in a range of macromolecular applications [56,57]. Many modern instruments do not rely on a TMS standard for sample calibration, so choices whether to rely on the direct solvent signal should be made based on the validity of the data and experimental/instrument design. The work of Li et al. in 2009 comment more specifically on how the diffusion of solvent signals vary with both viscosity and formula density [19]. A range of molecular liquids with known temperature dependent diffusions are reported for PFG measurements to assist instrument calibration [58].

In 2022 Voorter et al. derived a universal calibration, by combining terms in the previous equations with empirical Rouse-Zimm polymer models [11]. This is represented in Equation (9).

$$D\eta = AM^{-\alpha} \tag{9}$$

Here the parameter A negates the varying viscosity of the solvent in the presence of polymers [14]. This has proven successful in overcoming variations in sample molar mass and dispersity [11]. Further work by Ruzicka et al. has shown that varying pulse sequences to suit different solvent/polymer combinations led to improvements in data reliability and accuracy at higher molar masses [14].

There are also instrumentational considerations in the experiment design. As the signal decay time, and therefore the number of data points, varies between sample constituents; it is not possible to fully optimize the system with respect to viscosity and diffusion of solvated polymers with one measurement. For this study some pre-optimised settings were employed which adequately characterize many samples, giving fast and efficient gradient decays in a short time period. Unoptimized data is reflected in the error associated with the fitted diffusion parameter automatically generated by the fitting algorithm and provided sufficient points along the decay are measured the value will be accurate with a reasonable degree of certainty [59]. In order to study as wide a variety of diffusions as possible a linear gradient ramp was employed, as opposed to a quadratic ramp which is more useful for studying small ranges of known samples of a similar size.

Previous methods of using DOSY to gain information about polymer sizes have involved comparison of sample diffusions against calibration standards of known molar mass [10], effectively mirroring the calibration method used for size exclusion chromatography, which also separates via hydrodynamic radius. It has been shown previously that diffusion NMR can be used to indicate dispersity of samples because the signals from the chain ends are weighted according to the number of molecules whereas the signals from the repeat units are influenced by the mass of the chains and will tend towards M_w as the molar mass increases [60]. For the analysis of small analytes DOSY protocols have been become commonplace that recommend 16 gradient steps however this appears to be inadequate to represent the complexity of size distributions that many polymers present [3]. Work by Guest et al. has discussed the need to balance signal-to-noise ratios against instrument time for small molecules [61], although the dispersity of functional groups through polymer size distributions (which are themselves disperse) can introduce additional complexity in polymer analytes which will affect diffusion resolution [62].

Finally it is worth considering that diffusion measurements can be affected by internal convection within the NMR tubes, particularly when studying at elevated temperatures [63]. Studies have shown that this is less problematic for samples in deuterium oxide than for organic solvents such as chloroform or methanol [63]. All measurements in this paper were carried out at 25 °C, slightly above room temperature to assure uniformity between measurements. Samples were prepared to be at a constant concentration although as the sample viscosity is measured, D should be independent of exact concentration in dilute samples (this is something we have investigated in our supporting data - see ESI Section 8). Gross changes to the sample properties will affect both the diffusion of the solvent and the polymer peaks and so will self-negate during data interrogation. Studies of aqueous polymer solutions studied measured via variable temperature NMR have shown that employing in sample viscosity correction allowed for the sample temperature to be heated up to 35 °C before thermally induced convection caused the linearity of viscosity response to deviate [62]. In other solvents the issue of convection can be minimized by employing low-volume samples, or sample tubes with smaller horizontal dimensions [63]. Anyone designing experiments should be aware that vertical dimensions can also play a significant role in diffusion measurements as under/over filled samples can disrupt gradient shimming processes, whilst increasing sample volume could also impact temperature gradients. For this study we have elected to study samples at 0.8 mL, as for large polymer materials, we have found this volume offers ideal gradient shimming potential with the aforementioned broad temperature range [62].

All of the above considerations are necessary to consider in experimental design and they impact the treatment of polymeric diffusion data. As a result one of the reoccurring issues of DOSY NMR is the calibration of the instrument to provide a full decay profile for the analyte proton peaks [3]. There are many variables including gradient pulse shape and length, magnet strength and allowed relaxation times. In recent years newer diffusion based probes have been developed which can effectively target these slow diffusing species and provide full signal decay profiles where previously only partial decays could be observed. Ideally the user should be looking for a 2-log reduction in signal intensity across 6 or more gradient steps in order for the instrument to provide an accurate peak diffusion value. However, when a sample is analysed that contains two species of wildly divergent size (such as a solvent and a polymer) either a large number of gradient steps will be required to capture both or best practice would be to run two separate diffusion measurements, one to determine D_{poly} (D_p) and one to determine $D_{solvent}$ (D_S). Regardless of approach the size of the analytical parameters previously defined in combination with the Δ and δ from the instrument will determine the resultant gradient intensity decay profile. Without altering these parameters interrogating polymers of different size will show that there is both an upper and lower limit to the measurable size range of polymer materials (Fig. 1A). The exponents of the magnetic gradient (from Equation (2)) are summarised as B (with units Sm^{-2}), which is presented as the typical X axis of a Stejskal-Tanner plot to indicate the analyte signal intensity reduction across the measurement variables. This is, for DOSY measurements, an equivalent concept to the calibration range of a size exclusion chromatogram where the pore size limits the analytical range of the instrument (Fig. 1B).

1.3. Experimental purpose

The aim of this study is to combine all of the above to demonstrate the application of DOSY NMR across a broad selection of polymer molar masses with minimal experimental error. We are particularly interested in the advantages for increased instrument power, as reliably ascertaining diffusion values of slow moving species requires increasing the strength and pulse rate of the magnetic field to achieve the desired signal strength reductions. We will also investigate whether the assumptions outlined in Equations (7) and (8) are universal across different solvent/polymer/size combinations as they have previously only been shown in specific model systems [43] and never previously rigorously interrogated.

2. Materials and methods

Low dispersity polymer standards of known molecular weight were provided by Agilent. These standards, varying from 600 to 11.7 million g mol^{-1} were analysed using automated DOSY NMR spectroscopy, using a system designed to give acceptable signal intensity decays allowing an accurate diffusion value to be quoted for both the solvent (small molecule) and polymer backbone (large macromolecule) proton peak.

Deuterated NMR solvents were purchased from Cambridge Isotope Laboratories Inc. USA. High precision field matched NMR tubes were purchased from Wilmad USA. Most samples dissolved readily in solution however some (particularly PEG-Methanol) required gentle warming to 50 °C to encourage miscibility. NMR measurements were taken primarily on a Bruker Avance Neo 600 MHz equipped with either a DiffBB Probe or iProbe. Samples analysed with a DiffBB probe were initiated using Bruker Diff5 command to modify pulse sequence, targeting slowly diffusing signals whilst ensuring the gradient power dissipation was retained below the maximum threshold of the maximum instrument duty cycle. Additional measurements (presented in the Electronic Supporting Information, Sections 3, 4 and 9) were recorded on a Bruker Avance III HD 400 equipped with a SMARTprobe spectrometer operating at 400.13 MHz. Temperature accuracy for the ^{1}H DOSY experiments performed was ensured by initial calibration of the sample temperature to displayed sample control temperature by measurement of the shift between the residual CH_3 and OH resonances of methanol (99.8 % MeOD). Measurements were performed at 298.15 K.

Calibration of the gradient field strength was performed using a sample of H_2O in D_2O (1 % v/v) doped with $GdCl_3$ (0.1 mg) as a paramagnetic relaxation agent. Unless otherwise stated D are provided as M^2

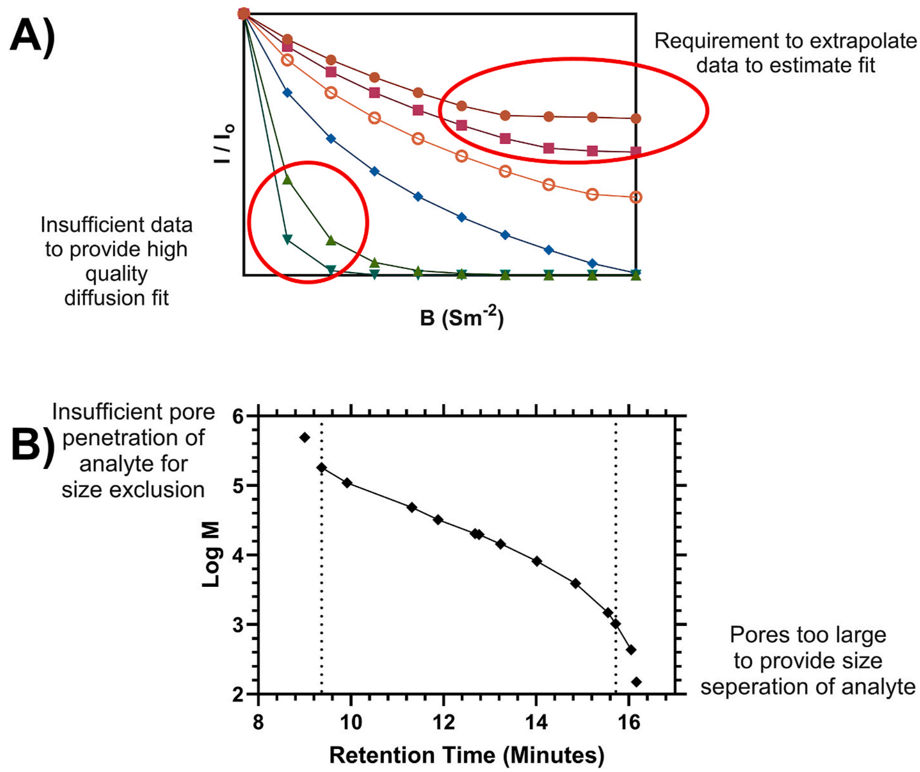

Fig. 1. Exemplar data showing modality of size analysis of Poly(ethylene glycol) standards (M_n 580–1,179,000 g mol^{-1}) via DOSY and SEC instruments. A) Target plot of proton signal decay across 10 gradient steps of DOSY measurement (Δ 0.01. δ 0.02). Data is reproduced with authors permission from Swift et al. [43]. b) Example indicate typical limits of column calibration type revealed by Size Exclusion Chromatography, the gold standard of polymer size analysis. PEG samples of known molar mass (M_p) elute at different times after passing through the stationary phase to expose the upper and lower exclusion limits of column porosity measurements. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

s^{-1}, δ and Δ are provided as seconds. All instruments using the DiffBB probe were preset with a Gamma (Gyromagnetic ratio of targeted nuclei) of 26,752 rad to target protons only, and a gradient calibration of 164.8816 G mm^{-1}. Gradient strengths were calibrated to provide a diffusion coefficient of 1.91×10^{-9} m^{-2} s^{-1} at 298.15 K (gradients operated from 95 % to 5 % using 16 points with a quadratic decay). The 600 MHz instrument fitted with a DiffBB probe is designed to allow for rapid variation of accurate pulse sequences to target specific nuclei, employing a typical settings of δ 0.01, Δ 0.04. Additional data collected using a Smartprobe on the 400 MHz magnet employed a bipolar LED sequence with typical δ 0.005 Δ 0.02 pulse widths, where gradient strengths incremented between 0.28 and 5.19 G mm^{-1} spaced equally across >16 steps.

Samples were prepared in D_2O at 1 mg mL^{-1} (unless stated otherwise), with brief heating to 60 °C to encourage complete dissolution. NMR tubes were filled with 0.8 mL solution to a constant volume. One dimensional ^1H experiments were recorded using 16 scans. Data analyses were performed using TopSpin software version 4.1.1.

To ensure that convection was not causing data aberrations a series of calibration measurements were undertaken using STE and dSTE pulse sequences – which provided evidence that the observed polymer diffusion varied by < 2 %. This data is shown in the electronic supporting data.

Diffusion values from these experiments were determined using Bruker Dynamic Centre module. Any gradient spectra showing errors (i.e. peak signal oscillating at high magnetic gradients due to low peak intensity) were discounted from the fit. Exemplar data on data fitting is shown in the ESI. Statistical analysis and data plotting was carried out in Graphpad Prism (V6.07) using a log-log nonlinear regression to determine best fits of data and determine outliers.

3. Results

3.1. Initial sample analysis

A blank sample of D_2O was scanned and gave a peak average diffusion value of 1.914×10^{-9} m^2 s^{-1} which is close to previously measured values for D_2O/H_2O mixtures [64,65]. This value was repeated several times (n = 3) to ensure accuracy and was found to vary by less than 0.1 %. The value will alter dependent on the ratio of H_2O/D_2O and as such should be determined experimentally with each instrument/sample combination.

Polymer samples were characterized both by size-exclusion chromatography and light scattering to give accurate M_p, M_n, M_w, M_z and Đ information (see supporting electronic information). The ^1H NMR of poly(ethylene oxide) in D_2O gives an NMR spectrum with only 1 proton peak arising from the hydrophobic polymer backbone (3.63 ppm) at a distinct position from residual protons in the solvent resonance (4.70 ppm). Some examples of raw data of one of the larger polymers studied (M_n 668,500 g mol^{-1}) with varying pulse sequences that target slowly diffusing species are shown in Fig. 2. Here it is seen, that when targeting extremely slow diffusing species (such as large molar mass polymers) very small variations in δ (as low as 20 μs) can produce significant variation on the quality of the results. In the Stejskal-Tanner plot (Fig. 2 A) the X axis is defined as B – which is the gradient strength (the exponential component of equation (2)), and so the slope of the regression line provides the diffusion coefficient. The stated lower limit of diffusion that the DiffBB probe can resolve is 10^{-13} M^2 s^{-1}, so no samples within this study approach the instrument boundaries. Further comparison of data outputs for these samples using both 400 and 600 MHz instruments are provided in the electronic supporting information (see ESI Section 3).

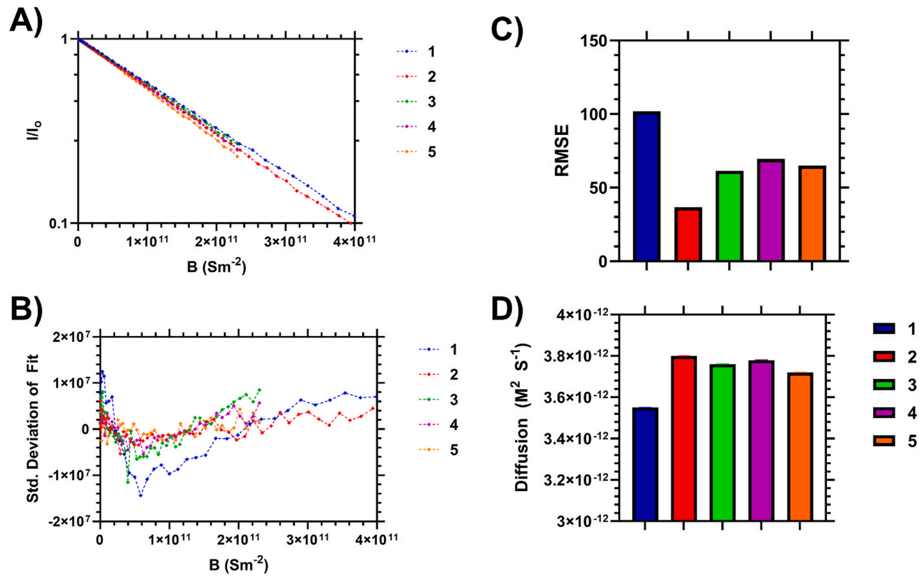

Fig. 2. Analysis of 668,500 g mol^{-1} PEO polymer sample using DiffBB probe with 5 different experimental parameters. **1**: δ 0.00102 s, Δ 0.04 s, I_0 5.68 × 10^8, RMSE 102.**2**: δ 0.00102 s, Δ 0.02 s, I_0 2.95 × 10^8, RMSE 36.**3**: δ 0.001 s, Δ 0.04 s, I_0 5.73 × 10^8, RMSE 61.**4**: δ 0.001 s, Δ 0.03 s, I_0 3.00 × 10^8, RMSE 69.**5**: δ 0.001 s, Δ 0.02 s, I_0 2.60 × 10^8, RMSE 64. Data displayed A) Stejskal-Tanner plot of 668,500 g mol^{-1} PEO polymer using different pulse sequence attuned on the DiffBB probe. B) Residual fitting profiles comparing standard deviation of fit from raw data for each pulse sequence. C) Output D_p values for each fitting profile. D) RMSE of fit.

The amplitude decay of the high molar mass polymer with varying δ and Δ are shown in a Stejskal-Tanner plot in Fig. 2A. The signal amplitude of large molar mass polymers where δ was maintained at 0.001 did not achieve a full log reduction across the measurement, however this was achieved by increasing δ to 0.00102. Data indicate that varying Δ between 0.02 and 0.04 had a minor impact on signal strength reduction, but does have an impact on the quality of the applied fit used to calculate D. When δ was varied between 0.001 and 0.00102 and Δ varied from 0.02 to 0.04 the Stejskal-Tanner profile shows variability in the linearity of the apparent peak decay. This is profiled in the residual standard deviation of the fit (**2B**), and variation in the apparent diffusion value (**2C**) and Root-Mean-Squared Error (RMSE) of the fit (**2D**). Here it can be seen that the lowest RMSE fit tends to a higher diffusion value (3.8 × 10^{-12} M^2 s^{-1}) but all fits are within 0.25 × 10^{-12} m^2 s^{-1} below this value. This deviation in raw data integrity is not unexpected, however variability here will feed forwards into uncertainty over apparent R_H or any molar mass calculated from this measurement. It can be noted that often this deviation of the fit appears to scale with dispersity of the analyte polymer. Alternatively, for complex systems (such as Pluronic polymers) the decay lineshape could represent two different components of the polymer environment (i.e. free polymer vs polymer engaged in micelle formation) which can be separated via an inverse Lapace Transformation [66].

The peak diffusion coefficients of all PEG samples measured using the DiffBB probe (D_p, D_{H2O}) calculated using Equations (3)–(5), with in solution viscosity correction (Equations (7) and (8)) are shown in Table 1. Interrogation of the raw data shows that the decay of the signal intensity, for high molar mass samples (Table 1, samples 5–7) required use of pulse sequence settings only accessible using the DiffBB probe if we wished to ensure intensity decays below 25 % of the initial I_0 across the measurement. The fitting profile of the decay indicates that there is not a normal distribution of the decay – but this is the same for all decays and the goodness of fit (Root Mean Square Error (RMSE) of the fit for the DiffBB probe varied between 7 and 30. In almost all cases this was smaller on the 600 MHz instrument with the DiffBB probe compared to the to a comparable 400 MHz instrument without a DiffBB probe (see ESI Figure S3.2 and S3.3). In summary the RMSE was small in almost all cases for the 600 MHz instrument fitted with the DiffBB probe instrument (see supporting information for raw data). This data was used to fit the MHS equation (Equation (6)) which produced an α value for the Mark Houwink Sakurada equation of ≈0.6, which is comparable with the literature [43].

Table 1
Diffusion values for polymer standards and solvent used to calculate sample viscosity and hydrodynamic radius using data from high field instrument using DiffBB probe.[a]

Sample	M_n g mol^{-1}	D_{H2O} m^2 s^{-1} x 10^{-9}	D_P m^2 s^{-1} x 10^{-9}	η kg m^{-1} s^{-1}	R_H nm
1	580	2.055	0.3061	0.0010	0.70
2	3730	2.007	0.1102	0.0010	1.89
3	14,330	1.853	0.0487	0.0011	3.95
4	68,850	2.005	0.0195	0.0010	10.72
5	493,500	1.391	0.0050	0.0015	29.22
6	668,500	1.314	0.0038	0.0016	35.53
7	1,179,000	1.317	0.0027	0.0016	48.71

[a] Data determined using 400 MHz instrument is shown in the Electronic Supporting Information for comparison (see ESI Section 5).

3.2. The need for in-solution viscosity correction factors

The diffusion values were used to determine the hydrodynamic radii of all samples of PEG, with and without consideration of the solvent-in-solution viscosity (Equations (7) and (8)), and then compared to known values in the literature. The data are provided in Fig. 3. Comparisons in the datasets were given by compiling all available literature data into one logarithmic fit (N = 54, slope 0.5448 ± 0.005958, Y Intercept - −1.659 ± 0.02621, R^2 0.9938) and the 7 datapoints gathered from the DOSY measurements with solvent-in solution viscosity correction into another (N = 7, slope 0.5450 ± 0.006271, Y Intercept - −1.638 ± 0.03063, R^2 9993). Comparison of these slopes shows that the 2 fits were identical in both slope (P = 0.9722) and intercept (P = 0.2893) so no statistical difference could be seen between the DOSY, with the solvent-in-solution correction, method of determining hydrodynamic radii and other reported methods.

However the data in Fig. 3 also shows the same samples processed without viscosity correction. When the correction for sample viscosity was not applied (by measuring D_{H2O} and applying equations (6) and (7)) the hydrodynamic radii of the higher molar mass polymers ceased to

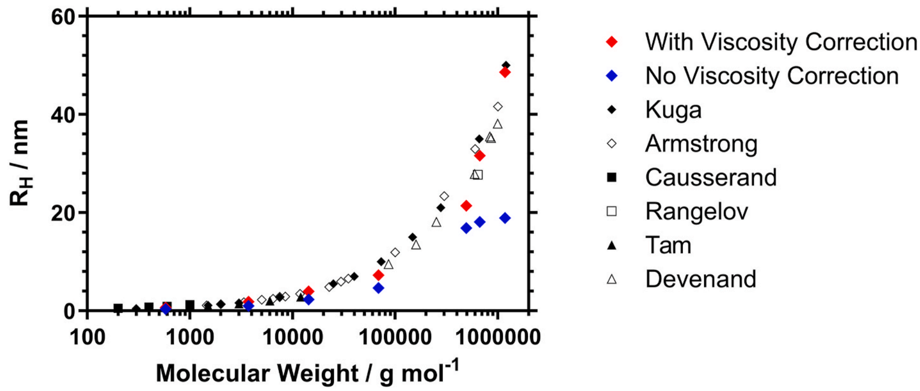

Fig. 3. Apparent hydrodynamic radius of PEO/PEG of varying molar mass calculated from NMR diffusion measurements with viscosity correction (red), without viscosity correction (blue), compared to literature values determined via light scattering and viscometric experiments [30,37,39–41]. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

scale with the molar mass once it approached 20 nm. Presumably at this point, and at this concentration, increases in PEG molar mass are coupled to viscosity increase of the solvent that must be taken account of. This resulted in an apparent 'diffusion limit' to the calculated polymer size. Reductions in sample viscosity would be achievable by consequently reducing the concentration further – but this increases the error of measuring reduced signal amplitudes as the peaks of the polymer sample become increasingly vulnerable to system background noise. An investigation into the effect concentration plays upon the increasing viscosity is shown in the electronic supporting information (Section 8) which indicates an upper concentration limit of 2 mg mL^{-1} regardless of the proposed solvent correction factor is applied. This is significantly lower than has been assumed in other recent studies using different polymer analytes/solvent combinations [15].

3.3. Sample/solvent matrix experiment

NMR analysis are typically carried out in a range of solvents and literature data for the hydrodynamic radii of polymers in other media are not so common. We fully analysed the potential of DOSY NMR across a range of polymer/solvent combinations. Soluble combinations of PEG, PMMA, PS and PNIPAM in D_2O, $CDCl_3$, MeOD, deuterated DMSO and deuterated acetonitrile (ACN) were studied. Results are shown in Figs. 4 and 6 which indicate that the determined hydrodynamic radii (leading to molar mass) for all sample/solvent/size combinations. Concurrently it can be assumed that intrinsic viscosities can also be predicted

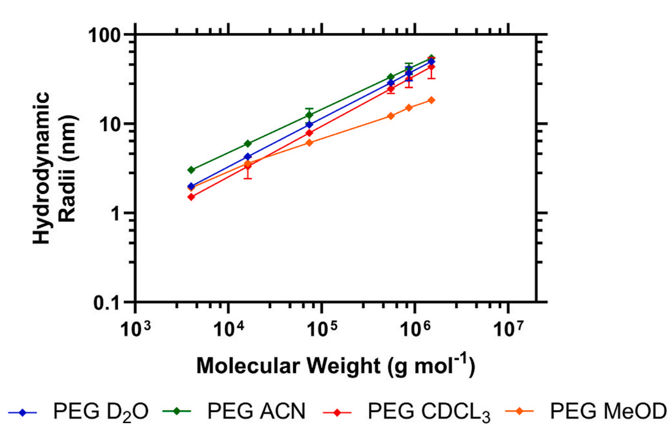

Fig. 4. Viscosity corrected hydrodynamic radii vs molar mass of PEG samples analysed in deuterated ACN (green, α = 0.46), D_2O (blue, α = 0.62), $CDCl_3$ (red, α = 0.70) and MeOD (orange, α = 0.52). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

following equations (3)–(5). As many size exclusion systems use the intrinsic viscosity of polymer standards to prepare a universal calibration it is important these can be accurately measured in exact solvent conditions to allow new protocols in analytical measurements to be developed [43].

To determine the instrumental error variability we carried out a matrix experimental design comparing hydrodynamic radii and intrinsic viscosities (using the solvent correction factor) with literature data where available (see supporting information ESI Section 9), with both 16 and 64 gradient pulses to allow us to account for instrument magnetic gradient strength, data point resolution, sample molar mass and solvent viscosity. Data recorded on the 400 MHz instrument is shown in the supporting information. The results of this study comparing determined hydrodynamic radii from our measurements to literature data trends are shown in Fig. 5 and further comparisons of this dataset with the closest comparable literature data using other analytical techniques are shown in the supporting information (Section 9).

The data showed that regardless of method, the apparent hydrodynamic radii scaled with the molar mass of the sample as expected as long as viscometric correction was applied. Table 2 shows there were clear differences in the observed error of the measurements depending on the number of diffusion steps recorded to allow for better gradient fitting. This was also repeated on the 400 MHZ instrument (see ESI Section 9) and overall larger error was observed there compared to the DiffBB probe setup. This was particularly evident for samples measured in methanol, the least viscous of the solvents analysed, which showed the greatest average error of all systems, particularly when using the 400 MHz (δ 0.001, Δ 0.02 gradient) system; indicating that for polymer analysis low viscosity can cause just as many issues as high viscosity. This correlates with the work of Swan et al. who determined a physical parameter (β) comparing inherent horizontal convection for these solvents at 25 °C [63]. Potentially this is due to increased convection in the sample which was partially overcome using the stimulated echo pulse sequence in the 600 MHz probe. Overall increasing the number of gradient steps from 16 to 64 reduced the average absolute % error on the t from 6.58 to 0.98.

An interesting observation is this data shows decreased % error in experimental measurements when the number of gradient steps were increased from 16 to 64. Given the paper discusses the analysis of single analytes with intense proton peaks the SNR is significantly diminished and thus this increase in the *N* of gradient steps should not be expected as necessary for accurately resolving *D*, as per the work on small molecules by Guest et al. [61] However – polymer peaks are broadened by both strong interactions with solvents and inhomogeneity in chemical structure (i.e. contain atactic chain structures) which means many overlapping signals contribute to the SNR. Additional work comparing the output diffusion value of a polymer sample looking to optimize the

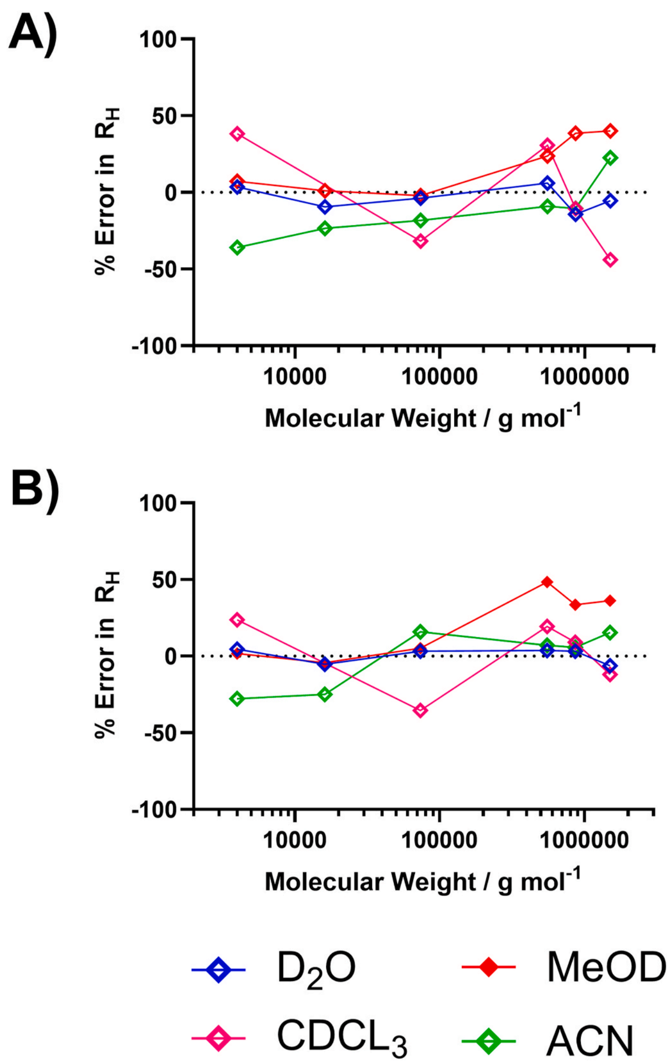

Fig. 5. % Error of determined R_H of PEO/PEG of varying molecular weights calculated from NMR diffusion measurements with a) 16 gradients and b) 64 gradients compared to literature measurements from previous studies (water [31–36], methanol [38], $CDCl_3$ [67], ACN [38]).

number of gradient steps, and the number of repeat scans per steps, to determine accurate D in $CDCl_3$ values is shown in the ESI Section 7. Here we found that at least 38 scans were necessary to resolve D for the analyte in $CDCL_3$. In this instance the desired resolution (and thus instrument run time) should potentially be considered against the dispersity of the analyte.

Further data are presented in Fig. 6 using PMMA/PS/PNIPAM polymers indicated that this technique is applicable across a range of polymer materials and solvent combinations. The % error for almost all solvent/polymer combinations was low (as is shown in the electronic supporting information in ESI Section 9), and was reduced further by increasing the number of gradient steps of the pulse sequence similarly to what was observed in Table 2 for the PEG materials.

4. Discussion

The results indicate that resolving the Stokes Einstein equation for two different components (solvent and polymer) in solution, can be used to determine solution viscosity in the presence of polymer. This can be used to accurately determine hydrodynamic radii of larger molar mass samples. The data correction enables the accurate measurement of polymers at a higher concentration and a higher molar mass than was previously possible. Depending on the experimental method used to determine D and η it has commonly been seen in the prior literature that the Stokes-Einstein equation can break down, with probes exhibiting significantly faster diffusion than expected from the macroscopic solution viscosity [68]. That the correction factor employed in this study at least partially overcomes this obstacle requires consideration in future work.

We have demonstrated that it is inappropriate to use the viscosity of the solvent, in the absence of the solute, for measuring polymer size via diffusion and it is necessary to use the solvent-in-solution viscosity. We have determined this by measuring the changing diffusion of the solvent experimentally to provide η of the solvent in the presence of the polymer. According to the polymer/solvent free volume theories of Duda et al. For a polymer/solvent binary mixture the diffusion of the solvent in the solution (D_{1s}) can be calculated from the diffusion of the pure solvent (D_s) via Equation (10).

$$D_{1S} = D_s(1 - \varnothing)^2(1 - 2x\varnothing)^2 \quad (10)$$

where $\varnothing$ represents the solute volume fraction and x is the polymer solvent interaction parameter. However if we considered the case where we approach the lower limit of concentration $\varnothing$ will tend to zero, indicating it is appropriate to consider D_{1S} as equal to D_S. This explains the accurate results previously reported in the literature for extrapolating molar mass of low concentration, low molar mass polymers from diffusion experiments, but this will not hold true for high molar mass

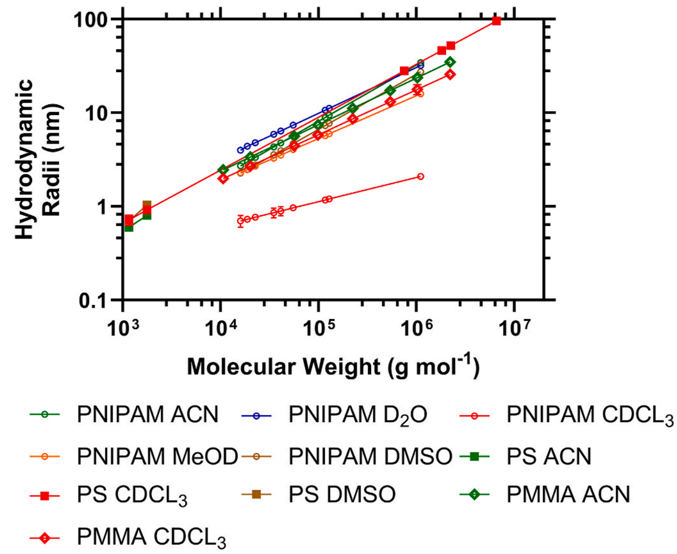

Fig. 6. Viscosity corrected hydrodynamic radii vs molar mass of PNIPAM, PS and PMMA samples analysed in deuterated ACN (green), D_2O (blue), $CDCL_3$ (red), MeOD (orange) and deuterated DMSO (brown). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Average absolute % error in apparent Hydrodynamic radii of PEG polymers compared to literature data trends varying the number of gradient steps across the DOSY measurement (i.e. measurement resolution).

	16 steps	64 steps
D_2O	3.88	0.51
MeOD	*18.1*	*20.2*
$CDCL_3$	3.44	0.96
ACN-D 3	12.4	1.47
Average[a]	6.58	0.98
Std. Dev[a]	5.06	0.47

[a] MeOD data discounted from average and Std. Dev calculations.

polymers that increasingly exhibit a greater impact on solvent volume fraction at the concentrations required to observe a distinct NMR signal. Therefore the DiffBB probe extends the potential range of molar masses that are viable for study with this technique, provided in solution solvent correction is applied to data treatment.

This work shows both the sample preparation and instrument impact on data validity. It is clear that the two probes offer distinct outcomes regarding data integrity with respect to viscosity and therefore, likely convection. The data emerging from the diffusion optimised DiffBB probe, with resolvable PEG + Solvent proton signal intensity decay at all molar masses, shows an important improvement in our understanding of analyte solvation compared to previous work done on low field instruments [43]. Typically, the technique is most commonly used for differentiating complex samples containing multiple components, but in the field of polymer science there is much interest in using it to determine the 'size' of polymer chains or aggregates. Whilst low-cost NMR instruments are becoming more widespread the work undertaken in this report shows the benefits of dedicated high field instruments with dedicated probes. The data shown in this report shows that higher field and gradient strength capable instruments provide distinct advantages in disclosing the high molar mass/high viscosity samples that exhibit slow diffusions – and whilst this is possible for low field instruments we anticipate it comes at a cost to data reproducibility. Furthermore the errors seen for low viscosity solvents, such as MeOD, where sample convection is more commonly observed, show this is not purely an issue for larger polymer samples but smaller, oligomeric, low viscosity materials as well. We do note that for this study we maintained a constant sample volume and tube diameter for comparison of data – experimental parameters that could be reduced to minimise convection issues common for these solvents. In future analysis alternative double stimulated pulse sequences designed to overcome any potential convection [16].

Applying a solvent correction factor to consider the sample viscosity allows for us to remove the previously observed 'diffusion limit', a problem common in other sizing technology for polymers. There remains a high potential for error for the study of high molar mass, highly viscous systems which demonstrate slow diffusion system if the probe used is not calibrated adequately. If accurate sizing of polymers is desired users should beware simply quoting 'diffusion' values apparent from their data without confirming the reduction in signal amplitude, ideally quoting the δ, Δ used in their pulse sequence and potentially indicating the RMSE of the fit.

5. Conclusions

This study has been designed to establish that two important concepts are essential to interpreting diffusion NMR data of polymer solutions. Firstly, that solvent correction must be applied to accurately extrapolate the hydrodynamic radii of dissolved polymer chains from apparent diffusion values. Secondly it explores the instrument variability that become increasingly apparent on high molar mass (high viscosity) samples. Use of a DiffBB probe on a 600 MHz instrument produced highly reproducible data with a close fit to the literature values even for high molar mass polymers.

CRediT authorship contribution statement

Thomas Swift: Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Edward Dyson:** Investigation, Formal analysis, Data curation. **Natalia Koniuch:** Investigation. **Richard Telford:** Writing – review & editing, Resources, Methodology, Funding acquisition. **Stephen Rimmer:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Stephen Rimmer reports financial support was provided by University of Bradford. Edward Dyson reports financial support was provided by University of Bradford. Richard Telford reports financial support was provided by University of Bradford. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.aca.2025.343937.

Data availability

Data will be made available on request.

References

[1] G. Pagès, V. Gilard, R. Martino, M. Malet-Martino, Pulsed-field gradient nuclear magnetic resonance measurements (PFG NMR) for diffusion ordered spectroscopy (DOSY) mapping, Analyst 142 (2017) 3771–3796, https://doi.org/10.1039/C7AN01031A.

[2] K.F. Morris, C.S. Johnson, Diffusion-ordered two-dimensional nuclear magnetic resonance spectroscopy, J. Am. Chem. Soc. 114 (1992) 3139–3141, https://doi.org/10.1021/ja00034a071.

[3] P. Groves, Diffusion ordered spectroscopy (DOSY) as applied to polymers, Polym. Chem. (2017), https://doi.org/10.1039/C7PY01577A.

[4] D.M. Meunier, Y. Li, W. Gao, Characterization of ultralarge polymers by gel permeation chromatography: challenges and opportunities, in: Recent Prog. Sep. Macromol. Part., American Chemical Society, 2018, pp. 6–89, https://doi.org/10.1021/bk-2018-1281.ch006.

[5] D.M. Meunier, J.H. Wade, M. Janco, R. Cong, W. Gao, Y. Li, D. Mekap, G. Wang, Recent advances in separation-based techniques for synthetic polymer characterization, Anal. Chem. 93 (2021) 273–294, https://doi.org/10.1021/acs.analchem.0c04352.

[6] D. Held, P. Kilz, Size-exclusion chromatography as a useful tool for the assessment of polymer quality and determination of macromolecular properties, Chem. Teach. Int. 3 (2021) 77–103, https://doi.org/10.1515/cti-2020-0024.

[7] E. Uliyanchenko, S. van der Wal, P.J. Schoenmakers, Deformation and degradation of polymers in ultra-high-pressure liquid chromatography, J. Chromatogr., A 1218 (2011) 6930–6942, https://doi.org/10.1016/j.chroma.2011.08.014.

[8] W. Li, H. Chung, C. Daeffler, J.A. Johnson, R.H. Grubbs, Application of 1H DOSY for facile measurement of polymer molecular weights, Macromolecules 45 (2012) 9595–9603, https://doi.org/10.1021/ma301666x.

[9] P. Lewinski, S. Sosnowski, S. Kazmierski, S. Penczek, l -Lactide polymerization studied by ^{1} H NMR with diffusion-ordered spectroscopy (DOSY): a "One NMR Tube Experiment" providing data on monomer conversion, polymer structure, M_n and M_w, Polym. Chem. 6 (2015) 4353–4357, https://doi.org/10.1039/C5PY00455A.

[10] A. Chen, D. Wu, J. Charles S. Johnson, Determination of molecular weight distributions for polymers by diffusion-ordered NMR, J. Am. Chem. Soc. 117 (1995) 7965–7970.

[11] P.-J. Voorter, A. McKay, J. Dai, O. Paravagna, N.R. Cameron, T. Junkers, Solvent-independent molecular weight determination of polymers based on a truly universal calibration, Angew. Chemie Int. Ed. n/a (2021), https://doi.org/10.1002/anie.202114536.

[12] F.M. Arrabal-Campos, P. Oña-Burgos, I. Fernández, Molecular weight prediction with no dependence on solvent viscosity. A quantitative pulse field gradient diffusion NMR approach, Polym. Chem. 7 (2016) 4326–4329, https://doi.org/10.1039/C6PY00691D.

[13] X. Guo, K.S. Wetzel, S.C. Solleder, S. Spann, M.A.R. Meier, M. Wilhelm, B. Luy, G. Guthausen, 1H PFG-NMR diffusion study on a sequence-defined macromolecule: confirming monodispersity, macromol, Chem. Phys. 220 (2019) 1900155, https://doi.org/10.1002/macp.201900155.

[14] E. Ruzicka, P. Pellechia, B.C. Benicewicz, Polymer molecular weights via DOSY NMR, Anal. Chem. 95 (2023) 7849–7854, https://doi.org/10.1021/acs.analchem.2c05531.
[15] O. Tooley, W. Pointer, R. Radmall, M. Hall, V. Beyer, K. Stakem, T. Swift, J. Town, T. Junkers, P. Wilson, D. Lester, D. Haddleton, MaDDOSY (mass determination diffusion ordered spectroscopy) using an 80 MHz bench top NMR for the rapid determination of polymer and macromolecular molecular weight, Macromol. Rapid Commun. (2024) e2300692, https://doi.org/10.1002/marc.202300692.
[16] I.W.F. Silva, A. McKay, A. Sokolova, T. Junkers, Towards the universal use of DOSY as a molar mass characterization tool: temperature dependence investigations and a software tool to process diffusion coefficients, Polym. Chem. (2024).
[17] W. Hiller, B. Grabe, The universal calibration for structure- and solvent-independent molar mass determinations of polymers using diffusion-ordered spectroscopy, Anal. Chem. 95 (2023) 18174–18179, https://doi.org/10.1021/acs.analchem.3c03797.
[18] P. Groves, M.O. Rasmussen, M.D. Molero, E. Samain, F.J. Cañada, H. Driguez, J. Jiménez-Barbero, Diffusion ordered spectroscopy as a complement to size exclusion chromatography in oligosaccharide analysis, Glycobiology 14 (2004) 451–456, https://doi.org/10.1093/glycob/cwh037.
[19] D. Li, G. Kagan, R. Hopson, P.G. Williard, Formula weight prediction by internal reference diffusion-ordered NMR spectroscopy (DOSY), J. Am. Chem. Soc. 131 (2009) 5627–5634, https://doi.org/10.1021/ja810154u.
[20] E.J. Cabrita, B. Stefan, DOSY studies of hydrogen bond association: tetramethylsilane as a reference compound for diffusion studies, Magn. Reson. Chem. 39 (2001) S142–S148.
[21] R. Neufeld, D. Stalke, Accurate molecular weight determination of small molecules via DOSY-NMR by using external calibration curves with normalized diffusion coefficients, Chem. Sci. 6 (2015) 3354–3364, https://doi.org/10.1039/C5SC00670H.
[22] N. Giuseppone, J.-L. Schmitt, L. Allouche, J.-M. Lehn, DOSY NMR experiments as a tool for the analysis of constitutional and motional dynamic processes: implementation for the driven evolution of dynamic combinatorial libraries of helical strands, Angew. Chem. Int. Ed. 47 (2008) 2235–2239, https://doi.org/10.1002/anie.200703168.
[23] J.K. Young, G.R. Baker, G.R. Newkome, K.F. Morris, C.S.J. Johnson, "Smart" cascade Polymers.1 modular syntheses of four-directional dendritic macromolecules with acidic, neutral, or basic terminal groups and the effect of pH changes on their hydrodynamic radii, Macromolecules 27 (1994) 3464–3471.
[24] R. Plummer, D.J.T. Hill, A.K. Whittaker, DOSY NMR studies of chemical exchange behavior of poly(2-hydroxyethyl methacrylate), Macromolecules 39 (2006) 3878–3889, https://doi.org/10.1021/ma0603550.
[25] X. Yan, F. Wang, B. Zheng, F. Huang, Stimuli-responsive supramolecular polymeric materials, Chem. Soc. Rev. 41 (2012) 6042–6065, https://doi.org/10.1039/C2CS35091B.
[26] H. Willcock, A. Lu, C.F. Hansell, E. Chapman, I.R. Collins, R.K. O'Reilly, One-pot synthesis of responsive sulfobetaine nanoparticles by RAFT polymerisation: the effect of branching on the UCST cloud point, Polym. Chem. 5 (2014) 1023–1030, https://doi.org/10.1039/C3PY00998J.
[27] M.A. van Dongen, B.G. Orr, M.M. Banaszak Holl, Diffusion NMR study of generation-five PAMAM dendrimer materials, J. Phys. Chem. B 118 (2014) 7195–7202, https://doi.org/10.1021/jp504059p.
[28] G. Assemat, B. Gouilleux, D. Bouillaud, J. Farjon, V. Gilard, P. Giraudeau, M. Malet-Martino, Diffusion-ordered spectroscopy on a benchtop spectrometer for drug analysis, J. Pharm. Biomed. Anal. 160 (2018) 268–275, https://doi.org/10.1016/j.jpba.2018.08.011.
[29] S.H. Maron, F.E. Filisko, Heats of solution and dilution for polyethylene oxide in several solvents, J. Macromol. Sci. Part B. 6 (1972) 79–90, https://doi.org/10.1080/00222347208224790.
[30] C.M. Tam, A.Y. Tremblay, Membrane pore characterization—comparison between single and multicomponent solute probe techniques, J. Membr. Sci. 57 (1991) 271–287.
[31] T.C. Amu, The unperturbed molecular dimensions of poly(ethylene oxide) in aqueous solutions from intrinsic viscosity measurements and the evaluation of the theta temperature, Polymer (Guildf). 23 (1982) 1775–1779, https://doi.org/10.1016/0032-3861(82)90121-5.
[32] D.M. Woodley, C. Dam, H. Lam, M. LeCave, K. Devanand, J.C. Selser, Draining and long-ranged interactions in the polyethylene oxide)/water good solvent system, Macromolecules 25 (1992) 5283–5286, https://doi.org/10.1021/ma00046a027.
[33] S. Kirinčič, C. Klofutar, Viscosity of aqueous solutions of poly(ethylene glycol)s at 298.15 K, Fluid Phase Equilib. 155 (1999) 311–325, https://doi.org/10.1016/S0378-3812(99)00005-9.
[34] S. Kawaguchi, G. Imai, J. Suzuki, A. Miyahara, T. Kitano, K. Ito, Aqueous solution properties of oligo- and poly(ethylene oxide) by static light scattering and intrinsic viscosity, Polymer (Guildf). 38 (1997) 2885–2891, https://doi.org/10.1016/S0032-3861(96)00859-2.
[35] M. Kawaguchi, M. Mikura, A. Takahashi, Hydrodynamic studies on adsorption of polyethylene oxide) in porous media. 2. Molecular weight dependence of hydrodynamic thickness, Macromolecules 17 (1984) 2063–2065, https://doi.org/10.1021/ma00140a032.
[36] J.K. Armstrong, R.B. Wenby, H.J. Meiselman, T.C. Fisher, The hydrodynamic radii of macromolecules and their effect on red blood cell aggregation, Biophys. J. 87 (2004) 4259–4270, https://doi.org/10.1529/biophysj.104.047746.
[37] S. Kuga, Pore size distribution analysis of gel substances by size exclusion chromatography, J. Chromatogr., A 206 (1981) 449–461, https://doi.org/10.1016/S0021-9673(00)88914-1.
[38] M.P.J. Dohmen, A.M. Pereira, J.M.K. Timmer, N.E. Benes, J.T.F. Keurentjes, Hydrodynamic radii of polyethylene glycols in different solvents determined from viscosity measurements, J. Chem. Eng. Data 53 (2008) 63–65, https://doi.org/10.1021/je700355n.
[39] C. Causserand, S. Rouaix, A. Akbari, P. Aimar, Improvement of a method for the characterization of ultrafiltration membranes by measurements of tracers retention, J. Membr. Sci. 238 (2004) 177–190, https://doi.org/10.1016/j.memsci.2004.04.003.
[40] S. Rangelov, W. Brown, Microgel formation in high molecular weight poly (ethyleneoxide), Polymer (Guildf). 41 (2000) 4825–4830, https://doi.org/10.1016/S0032-3861(99)00708-9.
[41] K. Devanand, J.C. Selser, Asymptotic behavior and long-range interactions in aqueous solutions of poly(ethylene oxide), Macromolecules 24 (1991) 5943–5947, https://doi.org/10.1021/ma00022a008.
[42] I. Avramov, Relationship between diffusion, self-diffusion and viscosity, J. Non-Cryst. Solids 355 (2009) 745–747, https://doi.org/10.1016/j.jnoncrysol.2009.02.009.
[43] T. Swift, R. Hoskins, R. Telford, R. Plenderleith, D. Pownall, S. Rimmer, Analysis using size exclusion chromatography of poly(N-isopropyl acrylamide) using methanol as an eluent, J. Chromatogr., A 1508 (2017) 16–23, https://doi.org/10.1016/j.chroma.2017.05.050.
[44] S. Pickup, F.D. Blum, Self-diffusion of toluene in polystyrene solutions, Macromolecules 22 (1989) 3961–3968, https://doi.org/10.1021/ma00200a025.
[45] N. Ramesh, P.K. Davis, J.M. Zielinski, R.P. Danner, J.L. Duda, Application of free-volume theory to self diffusion of solvents in polymers below the glass transition temperature: a review, J. Polym. Sci., Part B: Polym. Phys. 49 (2011) 1629–1644, https://doi.org/10.1002/polb.22366.
[46] J. Sharma, K. Tewari, R.K. Arya, Diffusion in polymeric systems–A review on free volume theory, Prog. Org. Coating 111 (2017) 83–92, https://doi.org/10.1016/j.porgcoat.2017.05.004.
[47] R.A. Waggoner, F.D. Blum, J.C. Lang, Diffusion in aqueous solutions of poly (ethylene glycol) at low concentrations, Macromolecules 28 (1995) 2658–2664.
[48] R.A. Waggoner, F.D. Blum, J.M.D. MacElroy, Dependence of the solvent diffusion coefficient on concentration in polymer solutions, Macromolecules 26 (1993) 6841–6848, https://doi.org/10.1021/ma00077a021.
[49] E.O. Stejskal, J.E. Tanner, Spin diffusion measurements: spin echoes in the presence of a time-dependent field gradient, J. Chem. Phys. 42 (1965) 288–292, https://doi.org/10.1063/1.1695690.
[50] C.S.J. Johnson, Diffusion ordered nuclear magnetic resonance spectroscopy: principles and applications, Prog. Nucl. Magn. Reson. Spectrosc. 34 (1999) 203–256, https://doi.org/10.1016/S0079-6565(99)00003-5.
[51] R. Hardy, R.L. Cottington, Viscosity of deuterium oxide and water in the range 5 to 125 C, J. Res. Natl. Bur. Stand. 42 (1949) 573, 1934.
[52] C. Strazielle, H. Benoit, Some thermodynamic properties of polymer-solvent systems. Comparison between deuterated and undeuterated systems, Macromolecules 8 (1975) 203–205, https://doi.org/10.1021/ma60044a021.
[53] G.H. Sørland, D. Aksnes, Artefacts and pitfalls in diffusion measurements by NMR, Magn. Reson. Chem. 40 (2002) S139–S146, https://doi.org/10.1002/mrc.1112.
[54] W. Li, G. Kagan, R. Hopson, P.G. Williard, Measurement of solution viscosity via diffusion-ordered NMR spectroscopy (DOSY), J. Chem. Educ. 88 (2011) 1331–1335, https://doi.org/10.1021/ed100941t.
[55] L. Van Lokeren, G. Maheut, F. Ribot, V. Escax, I. Verbruggen, C. Sanchez, J. C. Martins, M. Biesemans, R. Willem, Characterization of titanium dioxide nanoparticles dispersed in organic ligand solutions by using a diffusion-ordered spectroscopy-based strategy, Chem. Eur J. 13 (2007) 6957–6966, https://doi.org/10.1002/chem.200601722.
[56] M. Kaur, V. Yardley, K. Wang, J. Masania, A. Botana, R.R.J. Arroo, M. Li, Artemisinin cocrystals for bioavailability enhancement. Part 1: formulation design and role of the polymeric excipient, Mol. Pharm. 18 (2021) 4256–4271, https://doi.org/10.1021/acs.molpharmaceut.1c00384.
[57] E.R. McCarney, C.J. Breaux, P.M. Rendle, Measurement of the hydrodynamic radii of PEE-G dendrons by diffusion spectroscopy on a benchtop NMR spectrometer, Magn. Reson. Chem. 58 (2020) 641–647, https://doi.org/10.1002/mrc.4997.
[58] M. Holz, S.R. Heil, A. Sacco, Temperature-dependent self-diffusion coefficients of water and six selected molecular liquids for calibration in accurate 1H NMR PFG measurements, Phys. Chem. Chem. Phys. 2 (2000) 4740–4742, https://doi.org/10.1039/B005319H.
[59] R. Huo, R. Wehrens, J. va Duynhoven, L.M.C. Buydens, Assessment of techniques for DOSY NMR data processing, Anal. Chim. Acta 490 (2003) 231–251, https://doi.org/10.1016/S0003-2670(03)00752-9.
[60] J.R.D. Montgomery, C.S. Lancefield, D.M. Miles-Barrett, K. Ackermann, B.E. Bode, N.J. Westwood, T. Lebl, Fractionation and DOSY NMR as analytical tools: from model polymers to a technical lignin, ACS Omega 2 (2017) 8466–8474, https://doi.org/10.1021/acsomega.7b01287.
[61] J. Guest, P. Kiraly, M. Nilsson, G.A. Morris, Signal-to-noise ratio in diffusion-ordered spectroscopy: how good is good enough? Magnes. Res. 2 (2021) 733–739, https://doi.org/10.5194/mr-2-733-2021.
[62] B.D. Monnery, V.V. Jerca, R. Hoogenboom, T. Swift, Polymer conformation determination by NMR spectroscopy: comparative diffusion ordered 1H-NMR spectroscopy of poly(2-ethyl-2-oxazoline)s and poly(ethylene glycol) in D2O, Polym. Chem. (2024), https://doi.org/10.1039/D4PY00505H.
[63] I. Swan, M. Reid, P.W.A. Howe, M.A. Connell, M. Nilsson, M.A. Moore, G.A. Morris, Sample convection in liquid-state NMR: why it is always with us, and what we can do about it, J. Magn. Reson. 252 (2015) 120–129, https://doi.org/10.1016/j.jmr.2014.12.006.

[64] S.T. Wang, P.M.T. Hansen, S.A. Barringer, Effects of sucrose and CMC on D2O diffusion in water, Food Hydrocoll. 12 (1998) 115–119, https://doi.org/10.1016/S0268-005X(98)00002-2.
[65] R. Mills, Self-diffusion in normal and heavy water in the range 1-45, J. Phys. Chem. 77 (1973) 685–688, https://doi.org/10.1021/j100624a025.
[66] M. Nilsson, B. Håkansson, O. Söderman, D. Topgaard, Influence of polydispersity on the micellization of triblock copolymers investigated by pulsed field gradient nuclear magnetic resonance, Macromolecules 40 (2007) 8250–8258, https://doi.org/10.1021/ma071302p.
[67] S. 240 S.R. RheoSense, 2420 Camino Ramon, Intrinsic Viscosity Analysis of Polystyrene in Chloroform, n.d.
[68] G.D.J. Phillies, The hydrodynamic scaling model for polymer self-diffusion, J. Phys. Chem. 93 (1989) 5029–5039, https://doi.org/10.1021/j100350a008.